

Suppression of the onset and progression of collagen-induced arthritis in rats by QFGJS, a preparation from an anti-arthritic Chinese herbal formula

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Abstract

QFGJS is an herbal preparation, and its pronounced effectiveness in treating adjuvant-induced arthritis (AIA) has been previously demonstrated. We herein aimed to confirm its anti-arthritic effect on collagen-induced arthritis (CIA) in rats. CIA was established in female Wistar rats with intradermal injection of type II bovine collagen at the base of the tail of animals. CIA rats were treated daily with oral administration of different doses of QFGJS beginning on the day of the induction of arthritis (day 0, the prophylactic treatment) or on the day after the onset of arthritis (day 13, the therapeutic treatment) until day 30. The results showed that prophylactic treatment with QFGJS significantly suppressed the onset of arthritis, and therapeutic treatment with QFGJS markedly reduced paw swelling and ESR levels even in the established CIA. Radiologic and histopathologic changes in the arthritic joints were also significantly reduced in the QFGJS-treated versus vehicle-treated rats. Moreover, the serum levels of pro-inflammatory cytokines TNF- α , IL-1 β , and IL-6 were markedly lowered in the QFGJS-treated rats. Hence, our studies demonstrate the quality, safety, and effectiveness of QFGJS as an anti-arthritic agent, which makes QFGJS a strong candidate for further clinical trials on rheumatoid arthritis (RA) patients.

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Keywords: QFGJS; Herbal medicine; Rheumatoid arthritis; Collagen-induced arthritis; Anti-arthritis; Pro-inflammatory cytokines

1. Introduction

Rheumatoid arthritis (RA) is a common chronic and systemic autoimmune disorder characterized by inflammation of the syn-

ovial joints and concomitant destruction of cartilage and bone; it involves a complicated pathogenesis, with pathological changes in multiple targets (Pincus et al., 1994; Gabriel, 2001). The pro-inflammatory cytokines, mainly tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and IL-6 produced by monocytes, macrophages, and synovial fibroblasts, are suggested to play an important role in the pathogenesis and disease progression of RA (Feldmann et al., 1996; Gravalles and Goldring, 2000). TNF- α , IL-1 β , and IL-6 are present in highly increased amounts in synovial fluid and joint tissues of RA patients (Feldmann et al., 1996), while anti-TNF, anti-IL-1, and anti-IL-6 therapies have been reported to be effective in the treatment of RA (Szekanecz et al., 1998; Maini and Taylor, 2000).

Drug therapy for RA is based on two principal approaches: symptomatic treatment with non-steroidal anti-inflammatory drugs (NSAIDs) and disease-modifying antirheumatic drugs (DMARDs) (Smolen and Steiner, 2003). However, most of the currently available drugs are primarily directed towards the

Abbreviations: AIA, adjuvant-induced arthritis; CFA, complete Freund's adjuvant; CIA, collagen-induced arthritis; CII, bovine type II collagen; CMC, carboxymethylcellulose; DMARDs, disease-modifying antirheumatic drugs; ELISA, enzyme-linked immunosorbent assay; ESR, erythrocyte sedimentation rate; GLP, Good Laboratory Practice; H&E, hematoxylin and eosin; HPLC, high-performance liquid chromatography; ICSH, International Council for Standardization in Haematology; IFA, incomplete Freund's adjuvant; IL, interleukin; MMPs, matrix metalloproteinases; MT, *Mycobacterium tuberculosis*; NSAIDs, non-steroidal anti-inflammatory drugs; PBS, phosphate-buffered saline; RA, rheumatoid arthritis; SD, Sprague–Dawley; S.E.M., standard error of mean; SPF, specific-pathogen-free; TNF, tumor necrosis factor; TPA, 12-*O*-tetradecanoylphorbol 13-acetate

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control of pain and/or the inflammation associated with joint synovitis, but do little to interfere with the underlying immunoinflammatory events, and consequently also do little to block the disease progression and reduce cartilage and bone destruction of joints. A systematic review of randomized placebo-controlled trials conducted recently demonstrates that the published evidence supports only the efficacies of nine agents in Western medicine, i.e., infliximab, cyclosporin, sulphasalazine, leflunomide, methotrexate, parenteral gold, corticosteroids, auranofin and IL-1 receptor antagonist, in decreasing radiological progression in RA (Jones et al., 2003). Nevertheless, a variety of problems exist with these drugs. For example, the use of methotrexate and leflunomide is impeded by their long-term side-effects and toxicity (Kremer, 1999), while cytokines antagonists, despite substantial efficacy and clinical improvement, entail high cost and incur hypersensitivity to medications and infections (Feldmann and Maini, 2001). Consequently, there is dramatically growing interest in herbal medicines among persons with RA and in the RA research community (Rao et al., 1999). In fact, herbal medicine is being widely used virtually around the world for treatment of rheumatic and arthritic diseases, and in recent decades considerable advances have been made in both clinical and basic research on the treatment of RA (Rao et al., 1999; Shin et al., 2003; Soeken et al., 2003; Chang et al., 2005; Park et al., 2005). Thus, herbal medicines constitute a potentially important avenue leading to novel therapeutic agents for RA that may not only prevent structural damage of arthritic joints caused by tissue and bone breakdown, but also be safe, relatively inexpensive, highly tolerated, and convenient for many patients.

QFGJS is a pharmaceutical preparation from an anti-arthritic Chinese herbal formula composed of five herbs, i.e., *Sinomenium acutum* Rehder and Wilson (family Menispermaceae) stem, *Aconitum carmichaeli* Debx. (family Ranunculaceae) processed root, *Curcuma longa* L. (family Zingiberaceae) rhizome, *Paeonia lactiflora* Pall. (family Ranunculaceae) root, and *Paeonia suffruticosa* Andr. (family Ranunculaceae) root-bark, all of which have well-established histories of use for treatment of rheumatic and arthritic diseases in China, Japan and other Asian countries (Chen and Chen, 2001). Our previous pharmacological studies showed that QFGJS could significantly and dose-dependently suppress the paw edema of rats induced by carrageenan or various inflammatory mediators (e.g., histamine, serotonin) and the ear edema of mice induced by arachidonic acid or 12-*O*-tetradecanoylphorbol 13-acetate (TPA) (Zhou et al., 2006). We recently also reported that QFGJS could significantly inhibit the disease progression of adjuvant-induced arthritis (AIA) in rats, and markedly retard cartilage and bone destruction in the inflamed joints of AIA rats (Cai et al., 2005). In addition, the high-performance liquid chromatographic (HPLC) fingerprints of different batches of QFGJS demonstrated well-controlled and consistent quality of the herbal preparation produced by the pilot manufacturing processes. An acute toxicity test in both mice and rats showed only very low toxicity of QFGJS (Cai et al., 2005). Thus, previous work on QFGJS encourages further study, and the current work aimed to confirm the anti-arthritic

effect of QFGJS using collagen-induced arthritis (CIA) in rats.

CIA is an experimental autoimmune-mediated polyarthritis that is well established in susceptible strains of rodents by immunization with type II collagen (CII), the major constituent protein of articular cartilage. Compared to other experimental arthritis models, CIA is demonstrated to more closely resemble human RA in terms of clinical, histological, and immunological features as well as genetic linkage (Trentham et al., 1977; Billingham, 1983; Rosloniec et al., 1994). In the present study, the CIA model was tested to be sensitive to indomethacin with expectation of the known anti-inflammatory response. Indomethacin is one of the classical NSAIDs and most commonly used as the reference drug for laboratory research and preclinical development of novel anti-arthritic agents (Conway et al., 1995; Hamada et al., 2000; Cai et al., 2005). Using the rat CIA model, in the present study we demonstrated that QFGJS: (i) had a favorable safety profile, (ii) dramatically suppressed the onset and disease progression of CIA in rats, (iii) significantly lowered the serum levels of pro-inflammatory cytokines TNF- α , IL-1 β , and IL-6, and (iv) markedly protected the affected joints against cartilage destruction and bone erosion.

2. Materials and methods

2.1. Preparation of QFGJS

QFGJS was prepared from five herbs used as followed: *Sinomenium acutum* Rehder and Wilson stem 12 g, *Aconitum carmichaeli* Debx. processed root 9 g, *Curcuma longa* L. rhizome 6 g, *Paeonia lactiflora* Pall. root 15 g, and *Paeonia suffruticosa* Andr. root-bark 9 g. The sources of all five herbs and the manufacturing process used in the current preparation of QFGJS were the same as described previously (Cai et al., 2005). Finally, 51.0 g of all of five herbs yielded 5.83 g of QFGJS. For administration in all experiments, QFGJS solutions were prepared by dissolving the manufactured granules, in 0.3% (w/v) carboxymethylcellulose (CMC).

2.2. Animals

Sprague–Dawley (SD) and female Wistar rats (6–7 weeks old) were purchased from Laboratory Animal Services Center, the Chinese University of Hong Kong (Hong Kong, China). Animals were housed four per cage in rooms maintained at $20 \pm 1^\circ\text{C}$ with alternating 12-h light–dark cycle. Food and water were provided *ad libitum* throughout the experiments. Animals were acclimated to their surroundings over 1 week to eliminate the effect of stress prior to initiation of the experiments. All of the experimental protocols involving animals and their care were approved by the Committee on Use of Human & Animal Subjects in Teaching and Research of the Hong Kong Baptist University and were carried out according to the regulations of the National Institutes of Health of USA and the Department of Health of Hong Kong Special Administrative Region.

2.3. General toxicity studies of QFGJS

To generally detect potential long-term, repeat-dose toxicity of QFGJS and thereby define the characteristic, extent, dose- and time-dependent relationship, and target organs and tissues of potentially toxic effects, 92 male and 92 female rats were assigned randomly and evenly to 4 experimental groups. Rats were exposed to once-daily oral administration of QFGJS at doses of 0, 3.89, 6.80, and 9.72 g/kg (the four groups) for a duration of 3 months. Clinical signs (general behavior, fur condition, breathing and nose conditions, eyes and oral secretions, urine and faeces), toxic reactions, and mortality were monitored daily after the initiation of QFGJS treatment. Body weight and food consumption were recorded once or twice a week. At the end of the experiment, all animals were sacrificed and subjected to hematology and clinical chemistry tests, necropsy examination, organ weighing, and histopathologic examination. These toxicity studies were conducted in accordance with Good Laboratory Practice (GLP) Regulations for Non-clinical Laboratory Studies (Notification No. 2003-02) issued by the State Food and Drug Administration (SFDA) of China on 06 August 2003.

2.4. Induction of CIA and QFGJS treatment

CIA was induced in female Wistar rats using the method described previously (Rosloniec et al., 1994). Briefly, rats were intradermally injected at the base of the tail with 100 µg of bovine type II collagen (CII, Chondrex, Redmond, WA, USA) emulsified in incomplete Freund's adjuvant (IFA, Chondrex, Redmond, WA, USA). On day 7 after the primary immunization, all the rats were given a booster injection of 100 µg of CII in IFA.

Using the prophylactic treatment protocol beginning on the day of arthritis induction (day 0), CIA rats were treated once daily with oral administration of different doses of QFGJS (0.97, 1.94, and 3.89 g/kg body weight), indomethacin (Indo, Sigma, St. Louis, MT, USA) at the dose of 1 mg/kg body weight, and vehicle (0.3% CMC solution) until day 30 of the experiment. In the therapeutic treatment protocol for the established CIA, treatment with QFGJS, indomethacin, and vehicle were initiated on the day after the onset of arthritis (day 13) and continued once daily until day 30 of the experiment.

2.5. Evaluation of the development of arthritis

Rats were inspected daily for the onset of arthritis characterized by edema and/or erythema in the paws. The incidence and severity of arthritis were evaluated using a system of arthritic scoring, measurement of bi-hind paw volumes and body weight every 2 or 3 days beginning on the day when arthritic signs were first visible. Lesions of the four paws of each rat (i.e., the clinical arthritic signs) were graded from 0 to 4 according to the extent of both edema and erythema of the periarticular tissues; 16 was the potential maximum of the combined arthritic scores per animal (Rosloniec et al., 1994). The hind paw volumes were measured using a plethysmometer chamber (7140 UGO. Basile, Comerio, Italy), and expressed as the mean volume of both hind

paws of rat. The body weight of the rats was monitored with a 0.1 g precision balance (Sartorius AG, Goettingen, Germany).

2.6. Measurements of ESR and serum cytokine levels

Blood samples were collected from the arteries of the tails of animals on days 0, 10, 20, and 30, and erythrocyte sedimentation rate (ESR) was determined by a modified method based on International Council for Standardization in Haematology (ICSH) selected methods (Bull et al., 1993). Briefly, a 120 µl sample of blood was taken and directly dropped into 30 µl of 0.109 mol/L sodium citrate, mixed well, and then transferred into a 1.0 mm × 100 mm capillary tube (VWR International, West Chester, PA, USA). The tubes were held obliquely at an angle of 45 °C, and the results were recorded after 15 min.

Levels of the pro-inflammatory cytokines TNF-α, IL-1β, and IL-6 in blood serum were measured on days 20 and 23 for the prophylactic and therapeutic treatments, respectively, using commercially available ELISA kits for IL-1β (Pierce Biotechnology, Rockford, IL, USA), IL-6 and TNF-α (BD, San Diego, CA, USA) according to the manufacturers' recommendations.

2.7. Radiologic and histopathologic examinations

At the end of the experiments on day 30, rats were sacrificed by diethyl ether asphyxiation and the hind paws were radiographed on Fuji HR-Fast film (Fuji Photo Film, Tokyo, Japan), using a Giotto HT Mammography system (IMS, Bologna, Italy). Radiographs of each rat were evaluated for soft tissue swelling and bone erosion, and were scored blindly by two independent observers on a scale of 0 (normal), 1 (mild changes), 2 (moderate changes), and 3 (severe changes) (Nishikawa et al., 2003). The total radiologic scores were calculated from the sum of both hind paws, with a maximum possible score of 6 for each radiologic parameter per rat.

After X-ray check, the hind paws were fixed in 10% phosphate-buffered saline (PBS)-buffered formalin. The fixed tissues of the ankle joints were then decalcified in formic acid, embedded in paraffin, longitudinally cut into 5 µm sections, and stained with hematoxylin and eosin (H&E). Grading of cellular infiltration (polymorphonuclear cells, macrophages or lymphocytes), joint space narrowing, synovial hyperplasia, pannus formation, and bone and cartilage erosion of the ankle joints were examined by two independent observers using a semiquantitative scale from 0 (normal), 1 (mild changes), 2 (moderate changes), and 3 (severe changes) (Beehler et al., 2003; McCartney-Francis et al., 2003). Histopathologic scores were combined and expressed as the sum of both ankle joints, with a maximum histologic score of 6 for each histologic parameter per rat.

2.8. Statistical analysis

Data were expressed as the mean ± S.E.M. One-way analysis of variance with multiple comparisons using the Student–Newman–Keuls test was used to analyze differences between different groups. Incidence of arthritis was evaluated

by the χ^2 -test with multiple comparisons using Fisher's exact test. Statistical significance was accepted for $P < 0.05$.

3. Results

3.1. General toxicity of QFGJS

In the chronic oral toxicity studies, about 67% animals treated with QFGJS at the dose of 9.72 g/kg (equivalent to 100-fold human clinical dose) showed mild or moderate decrease of activity and diarrhea in the initial stage of the experiment (7–14 days after the initiation of QFGJS treatment). Significant loss of body weight of animals treated with doses of 3.89, 6.80, and 9.72 g/kg (equivalent to 40-, 70- and 100-fold human clinical doses, respectively) was observed after 6 weeks of daily administration of QFGJS (Fig. 1). The results of laboratory investigation showed that all hematological parameters (erythrocyte, leukocyte and platelet counts, hemoglobin concentration, hematocrit, lymphocyte and differential lymphocyte counts) and blood biochemical parameters (aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, total protein, albumin, total bilirubin, glucose, creatinine, triglycerides) of QFGJS-treated rats fell within the range of normal biological variation as compared with vehicle-treated rats (data available upon request). No mortality, specific toxic responses, and treatment-induced pathological changes in all test organs and tissues were observed in animals after 3 months of repeat dosing with QFGJS.

3.2. Effects of prophylactic treatment with QFGJS on the onset and disease progression of CIA

Rats treated with vehicle (i.e., control group) first displayed visible arthritic signs characterized by edema and/or erythema in paws around day 9 after the primary immunization, and showed maximum paw swelling by day 19, which gradually diminished thereafter (Fig. 3A and B). The total incidence of arthritis in

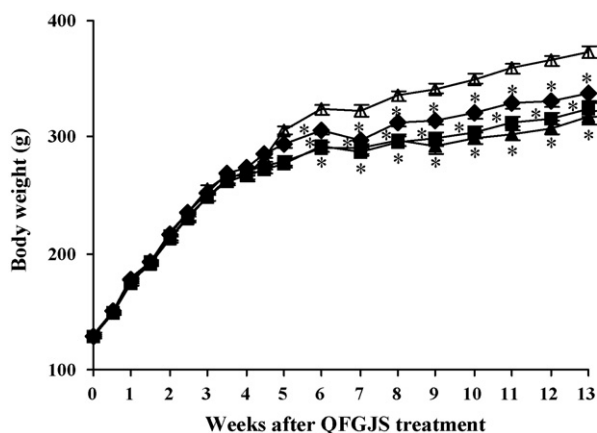


Fig. 1. Influence of QFGJS on body weight of rats in a 3-month, repeat-dose oral toxicity test. Rats were orally treated daily with QFGJS at doses of 0 (Δ), 3.89 ($\blacklozenge$), 6.80 ($\blacksquare$), and 9.72 ($\blacktriangle$) g/kg body weight for continuous 3 months. Data are expressed as mean $\pm$ S.E.M. ($n = 46$). * $P < 0.05$ vs. the vehicle-treated rats.

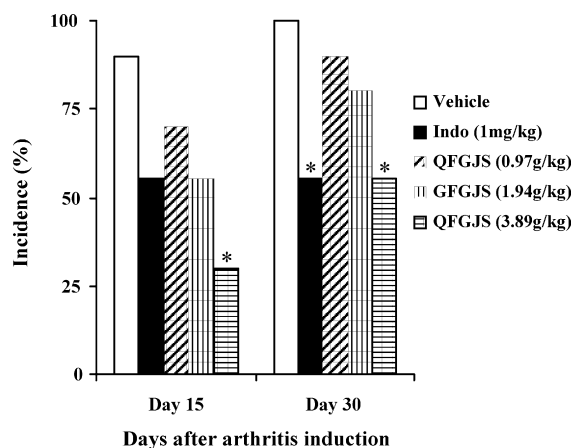


Fig. 2. Incidences of arthritis on days 15 and 30 in different groups of rats treated with QFGJS, indomethacin, and vehicle beginning on day 0 after the immunization with CIA. * $P < 0.05$ vs. the vehicle-treated group.

vehicle-treated rats reached 100% by day 17 (Fig. 2); while treatment with QFGJS beginning on the day of arthritis induction (day 0) significantly lowered the incidence of arthritis and markedly reduced paw swelling in a dose-dependent manner throughout disease progression of CIA (Figs. 2 and 3A and B). In addition, the development of CIA in vehicle-treated rats was accompanied by high elevation of ESR over the disease course, and treatment with QFGJS induced a significant and dose-dependent decrease of ESR levels (Fig. 3C).

3.3. Effects of therapeutic treatment with QFGJS on the disease progression of the established CIA

Following the primary and booster injections of CII/IFA, rats developed arthritis from day 10 onwards, and treatment with QFGJS, indomethacin, and vehicle were initiated on day 13 after the primary immunization and continued until day 30 of the experiment. As shown in Fig. 4A and B, even after the onset of arthritis, QFGJS at high doses (1.94 and 3.89 g/kg) did markedly reduce arthritic score and hind paw volume of the arthritic rats in a dose-dependent manner as compared to the vehicle-treated arthritic rats. Measurement of ESR showed that QFGJS at the dose of 3.89 g/kg significantly decreased the levels of ESR in CIA rats on day 20 (Fig. 4C).

3.4. Effects of QFGJS on radiologic changes of joint damage of the CIA rats

Representative radiographs of the hind paws from normal (grade 0), vehicle-treated (grade 2), and QFGJS-treated (grade 1) rats are shown in Fig. 5A–C, respectively. Radiologic analysis revealed severe soft tissue swelling and bone erosion in the joints of the vehicle-treated CIA rats (Fig. 5B) in comparison with the joints of normal rats (Fig. 5A). In contrast, QFGJS at the dose of 3.89 g/kg in both treatment protocols markedly inhibited soft tissue swelling and bone erosion of the arthritic joints, as did QFGJS at the dose of 1.94 g/kg only in the prophylactic, but not therapeutic, treatment protocol (Fig. 5C–E).

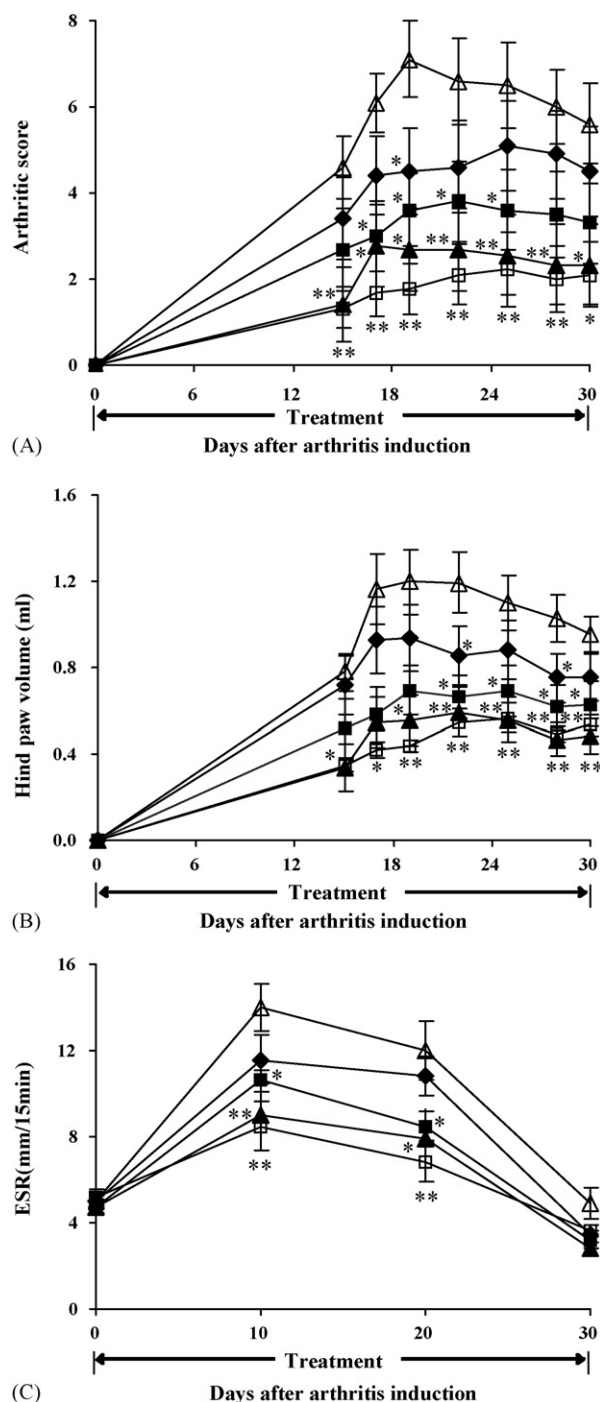


Fig. 3. Effects of prophylactic treatment with QFGJS on CIA in rats. Rats were orally treated daily with 0.97 (◆), 1.94 (■) and 3.89 g/kg (▲) QFGJS, 1 mg/kg indomethacin (□), and vehicle (Δ), beginning on day 0 after the immunization with CIA until day 30. Data were expressed as mean $\pm$ S.E.M. ($n=10-12$). * $P<0.05$; ** $P<0.01$ vs. the vehicle-treated arthritic rats.

3.5. Effects of QFGJS on histologic changes of joint damage of the CIA rats

Representative histologic sections of the ankle joints from normal (grade 0), vehicle-treated (grade 2), and QFGJS-treated (grade 1) rats are shown in Fig. 6A–C, respectively. Histopathologic examination of the synovial joints of the vehicle-treated

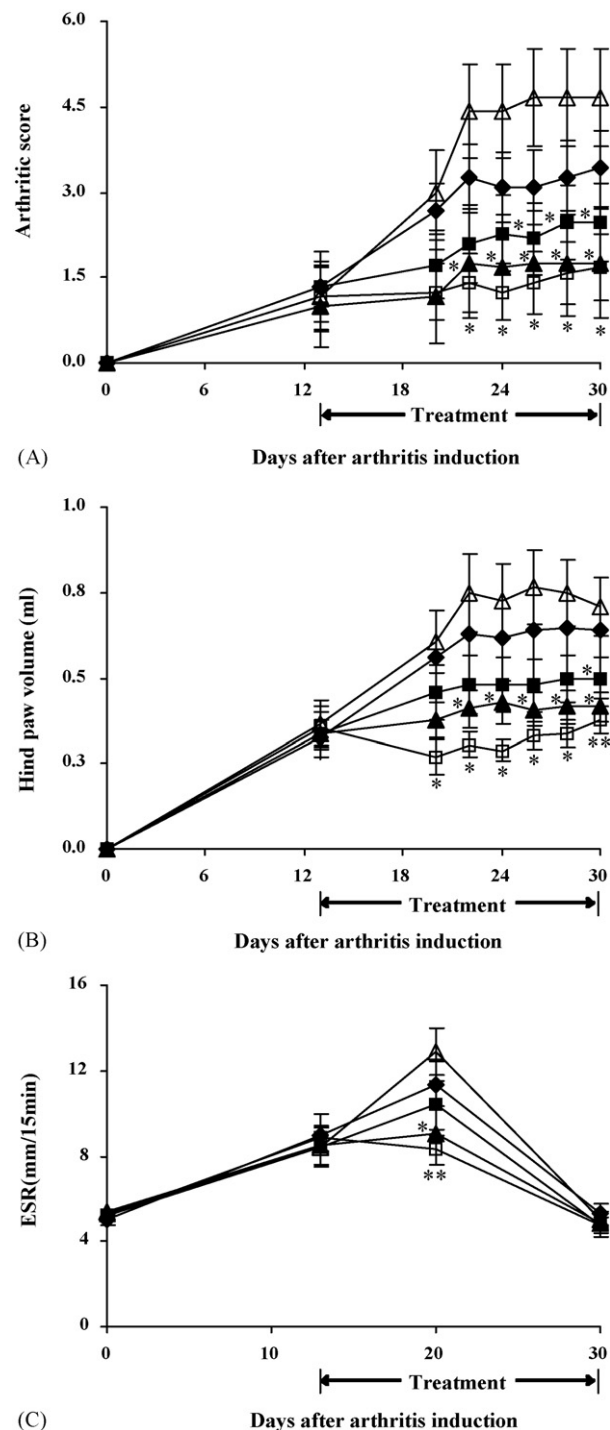


Fig. 4. Effects of therapeutic treatment with QFGJS on disease progression of the established CIA in rats. Rats were orally treated daily with 0.97 (◆), 1.94 (■) and 3.89 g/kg (▲) QFGJS, 1 mg/kg indomethacin (□), and vehicle (Δ), beginning on day 13 after the immunization with CIA until day 30. Data were expressed as mean $\pm$ S.E.M. ($n=10-12$). * $P<0.05$; ** $P<0.01$ vs. the vehicle-treated arthritic rats.

CIA rats revealed marked cellular infiltration, synovial hyperplasia, and joint space narrowing. Severe pannus formation and focal erosions of cartilage and bone in areas of direct pannus invasion were also observed in the synovial joints of vehicle-treated rats (Fig. 6B) as compared to the synovial joints of

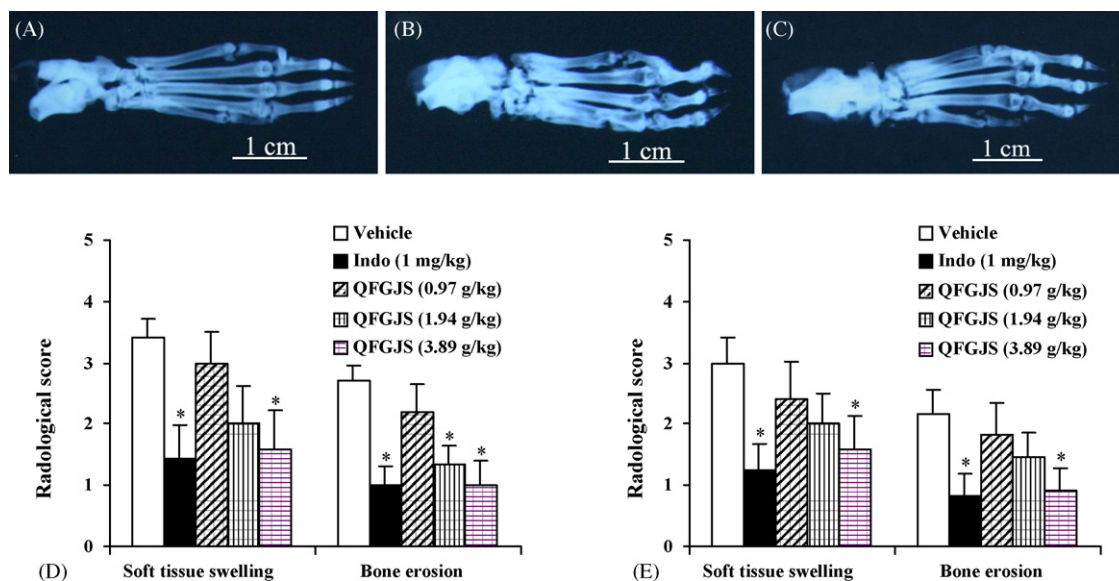


Fig. 5. Effects of QFGJS on radiologic changes of the CIA rats. Rats were orally treated daily with different doses of QFGJS, indomethacin, and vehicle, beginning on day 0 (D, the prophylactic treatment protocol) or day 13 (E, the therapeutic treatment protocol) after the immunization with CIA until day 30. Shown are representative radiographs of the hind paws from normal (A), vehicle-treated (B), and QFGJS-treated (C) arthritic rats. Severe soft tissue swelling and bone erosion in a representative arthritic hind paw (B) compared to a normal hind paw (A) are observed. These radiologic changes were markedly reduced in the QFGJS-treated rats compared to the changes of the vehicle-treated rats (C–E). Data are expressed as mean $\pm$ S.E.M. ($n = 10$ – 12). * $P < 0.05$; ** $P < 0.01$ vs. the vehicle-treated arthritic rats.

normal rats (Fig. 6A). In the QFGJS-treated rats, these pathological events in the ankle joints occurred to a markedly lesser degree (Fig. 6C–E).

3.6. Effects of QFGJS on serum TNF- α , IL-1 β , and IL-6 levels

Levels of pro-inflammatory cytokines TNF- α , IL-1 β and IL-6 in the blood serum were determined on days 20 and 23 for the prophylactic treatment beginning on day 0 and the therapeutic treatment beginning on day 13 after the immunization with CIA, respectively. Consistent with the joint swelling and the elevated levels of ESR, TNF- α , IL-1 β , and IL-6 in the vehicle-treated CIA rats were systemically overproduced in the serum; while the elevated TNF- α , IL-1 β , and IL-6 levels of the rats treated with QFGJS were decreased in a dose-dependent manner (Fig. 7). QFGJS at the dose of 3.89 g/kg only in the prophylactic, but not the therapeutic, treatment protocol significantly reduced the serum levels of TNF- α (Fig. 7A). In both prophylactic and therapeutic treatment protocols, QFGJS at doses of 1.94 and 3.89 g/kg markedly decreased the serum IL-1 β levels of the CIA rats (Fig. 7B). In contrast, markedly low serum levels of IL-6 were seen in the CIA rats either treated prophylactically with QFGJS at the dose of 3.89 g/kg or treated therapeutically with QFGJS at doses of 1.94 and 3.89 g/kg (Fig. 7C).

4. Discussion

RA is a complicated refractory autoimmune disease characterized by a number of the inflammatory and destructive events such as joint pain and swelling, synovial hyperplasia, pannus formation, cartilage and bone erosions, joint malforma-

tion (Weyand, 2000). Although various therapeutic approaches, such as cytokine-, receptor-, signal-transduction- or cell-type-directed intervention, have been done, the inflammatory and destructive processes of RA are impeded partly, but rarely blocked fully, which reflects both the complexity and multiplicity of the pathogenesis. These findings indicate that treatment targeting individual molecules might not suffice to stop the ongoing events, while combination therapy, which interferes with multiple targets, might be more effective for the treatment of RA (Smolen and Steiner, 2003). In this regard, herbal medicines, comprising multiple active components with broad-ranging pharmacological activities, represent an ancient model of combination therapy that could serve us well today.

QFGJS is a botanical drug product that are intended to treat RA and that is now waiting for the approval of the SFDA of China for clinical trials. QFGJS is formulated from five herbs with well-documented histories of human use. Such a formula is chemically quite complex, with hundreds if not thousands of constituents. However, several bioactive chemicals contained in the corresponding individual herbs in the formula have been identified, and the comprehensive information on their pharmacological activities relevant to anti-inflammation and anti-arthritis is readily available (Hikino et al., 1980; Bisset, 1981; Rafatullah et al., 1990; Liu et al., 1994, 1996; Suzuki et al., 1994; Tai and Hopkins, 1998; Tsai et al., 2001; Chou, 2003). Thus, QFGJS may able to provide a novel therapeutic intervention with multiple targets *via* various bioactive chemicals and the related pharmacological pathways.

The safety and effectiveness of herbal products are derived from well-controlled and batch-to-batch consistent quality of the production. In our previous qualitative and quantitative analyses, HPLC fingerprint profiles of different batches of QFGJS

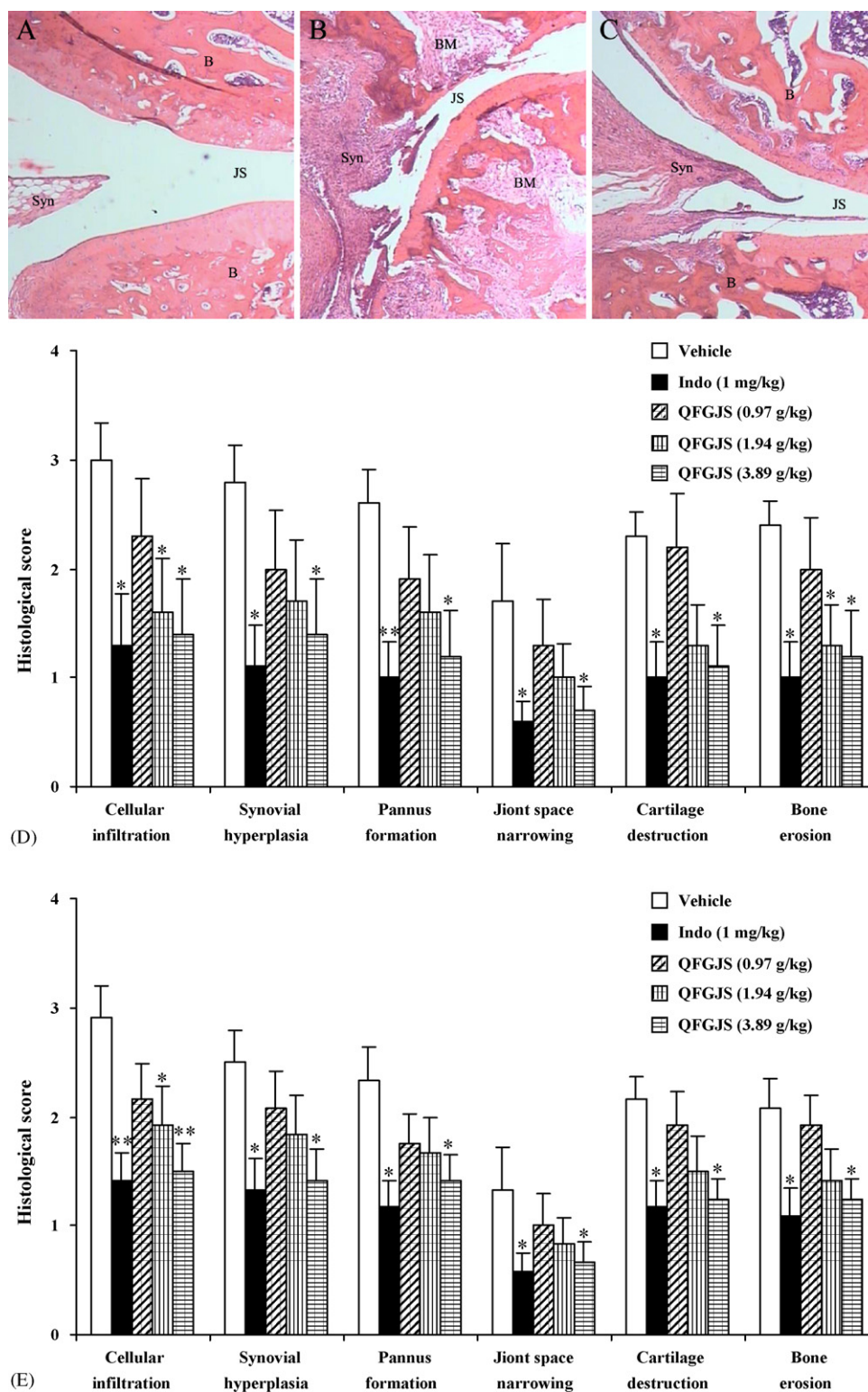


Fig. 6. Effects of QFGJS on histopathologic changes of the CIA rats. Rats were orally treated daily with different doses of QFGJS, indomethacin, and vehicle, beginning on day 0 (D) or day 13 (E) after the immunization with CIA until day 30. Shown are representative H&E stained (40 $\times$) histopathological sections of the tibiotarsal joints from normal (A), vehicle-treated (B), and QFGJS-treated (C) arthritic rats. Cellular infiltration, synovial (Syn) hyperplasia and pannus formation, narrowing of joint space (JS), and bone (B) erosion in the representative arthritic joint of the CIA rats (B) compared to a normal joint (A) were significantly observed, while these histopathologic changes were markedly reduced by treatment with QFGJS (C–E). Data are expressed as mean $\pm$ S.E.M. ($n = 10$ – 12). * $P < 0.05$; ** $P < 0.01$ vs. the vehicle-treated arthritic rats.

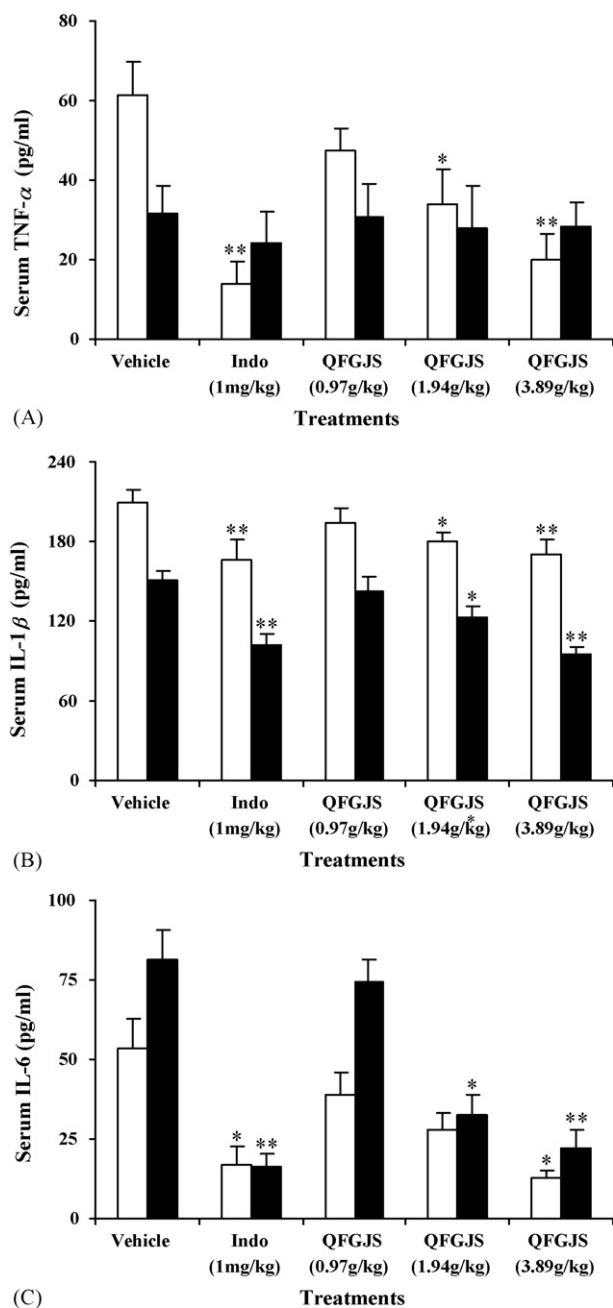


Fig. 7. Effects of QFGJS on serum TNF- α (A), IL-1 β (B), and IL-6 (C) levels of the CIA rats. Rats were orally treated daily with different doses of QFGJS, indomethacin, and vehicle, beginning on day 0 (□) or day 13 (■) after the immunization with CIA until day 30. Data are expressed as mean $\pm$ S.E.M. ($n=8-12$). * $P<0.05$; ** $P<0.01$ vs. the vehicle-treated arthritic rats.

have shown that the contents of five characteristic and bioactive chemical markers are relatively consistent. These marker are: sinomenine, present in the tem of *Sinomenium acutum* Rehder and Wilson; hyaconitine, present in the processed root of *Aconitum carmichaeli* Debx.; curcumin, present in the rhizome of *Curcuma longa* L.; paeoniflorin, present in the root of *Paeonia lactiflora* Pall.; paeonol, present in the root-bark of *Paeonia suffruticosa* Andr. (data were submitted for publication elsewhere). Furthermore, the long-term, repeat-dose

toxicological evaluation of QFGJS at doses of 3.89, 6.80, and 9.72 g/kg per day, administered continuously for 3 months, revealed no mortality nor serious toxicity in all rats treated with QFGJS, demonstrating that QFGJS has a favorable safety profile (Fig. 1).

The current study demonstrates that oral treatment with QFGJS can effectively block the disease progression of CIA, showing suppression of joint inflammation and of the radiologic and histopathologic progression of joint damage (i.e., it reduced cartilage destruction and bone erosion). Prophylactic treatment with QFGJS, beginning on the day of primary immunization (day 0), significantly inhibited the onset of CIA, while therapeutic treatment with QFGJS, beginning after the onset of arthritis (day 13), markedly suppressed disease progression of the established CIA, in a dose-dependent manner (Figs. 2–4). Both radiologic and histopathologic analyses revealed that, in relation to the development of inflammatory and arthritic lesions in the paws, most of the vehicle-treated CIA rats experienced progressive cartilage destruction and bone erosion, finally leading to complete ankylosis and malformation of the joints (Figs. 5 and 6). In contrast, both prophylactic and therapeutic treatments with QFGJS not only showed significant reduction of cellular infiltration, joint space narrowing, synovial hyperplasia, and pannus formation, but also markedly protected the affected joints against cartilage destruction and bone erosion (Figs. 5 and 6).

The inflammatory process in the CIA rats treated with vehicle was shown to lead to substantial increases in systemic levels of the pro-inflammatory cytokines TNF- α , IL-1 β , and IL-6; while QFGJS suppressed the overexpressed TNF- α , IL-1 β , and IL-6 in blood serum in a dose-dependent manner. It is believed that these pro-inflammatory cytokines help to propagate a local or systemic inflammatory process, and to induce biosynthesis and secretion of matrix metalloproteinases (MMPs) and osteoclasts that critically contribute to degradation of extracellular matrix and erosion of bone, respectively (Burger et al., 1998; Gravalles and Goldring, 2000; Smolen and Steiner, 2003). Our results indicated that QFGJS suppressed the polyarticular inflammatory process and joint damage by reducing the production of TNF- α , IL-1 β , and IL-6 (Fig. 7). The levels of pro-inflammatory cytokines TNF- α , IL-1 β , and IL-6 levels in blood serum were determined on days 20 and 23 for prophylactic and therapeutic treatment protocols, respectively. QFGJS significantly reduced the serum TNF- α level in the prophylactic treatment (Fig. 7A). In contrast, QFGJS could suppress the systemic overproduction of IL-1 β and IL-6 in both prophylactic and therapeutic treatments (Fig. 7B and C). Anti-TNF, anti-IL-1 or anti-IL-6 therapy has been proven to be clinically effective in RA (Szekanecz et al., 1998; Maini and Taylor, 2000). In a recent preclinical study, combination therapy with both IL-1 receptor and soluble TNF receptor antagonists is found to be more effective than either one alone administered in the treatment of CIA or AIA (Bendele et al., 2000; Feige et al., 2000). In the present study we found that QFGJS suppressed the overproduction of all cytokines TNF- α , IL-1 β , and IL-6. Therefore, orally applicable QFGJS may offer advantages and conveniences comparable to the combination of several anti-cytokine agents in the treatment of RA.

5. Conclusion

In the current study, QFGJS is demonstrated to not only inhibit the onset of arthritis by prophylactic treatment, but also to suppress the progression of soft tissue inflammation and joint destruction by therapeutic treatment on the established CIA in rats. QFGJS is an example of combination therapy, which, increasingly, is being found to be the best approach to complex refractory and degenerative diseases. The results obtained in both the present and recent previous studies provide substantial evidence of the quality, safety, and effectiveness of QFGJS as an anti-arthritic agent, while QFGJS is relatively inexpensive and convenient to administer (orally). Hence, QFGJS is a strong candidate for further clinical trials on RA patients.

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